

LAB # 02

Implementation of Scanner class for reading and taking inputs in java.

Objective

To get familiarity with the Scanner Class for reading and taking inputs in java.

Lab Goals

To demonstrate and gain familiarity with the Scanner class in Java for reading and taking inputs:

- a. **Understand Scanner Class Basics:** Gain a foundational understanding of the Scanner class in Java, including its purpose, syntax, and basic usage for reading input from various sources such as the console, files, or strings.
- b. **Practice Input Reading:** Learn how to use the Scanner class to read different types of input data, including integers, floating-point numbers, strings, and characters, and understand the methods provided by the Scanner class for retrieving this input.

PRINT FUNCTION IN JAVA

`java System.out.println()` is used to print an argument that is passed to it. The statement can be broken into 3 parts which can be understood separately as:

1. **System:** It is a final class defined in the `java.lang` package.
2. **out:** This is an instance of `PrintStream` type, which is a public and static member field of the `System` class.
3. **println():** As all instances of `PrintStream` class have a public method `println()`, hence we can invoke the same on `out` as well. This is an upgraded version of `print()`. It prints any argument passed to it and adds a new line to the output. We can assume that `System.out` represents the Standard Output Stream.

Syntax:

4. `System.out.println(parameter)`

Example

```
class demo {  
    public static void main(String[] args){  
        System.out.println("Welcome");  
        System.out.println("To");  
        System.out.println("ssuet");  
    }  
}
```

Output:

```
Welcome  
To  
ssuet
```

PRINT CONCATENATED STRINGS

```
classPrintVariables{  
    public static void main(String[] args){  
        Double number = -10.6;  
        System.out.println("I am " + "awesome.");  
        System.out.println("Number = " + number);  
    }  
}
```

Output:

```
I am awesome.  
Number = -10.6
```

SCANNER CLASS FOR I/O:

The Scanner class in Java is used for obtaining input from the user. It is part of the ***java.util package***. You can create a Scanner object and use its methods to read different types of input from the keyboard.

```
Scanner in = new Scanner(System.in);  
Scanner in = new Scanner("Hello
```

```
import java.util.*;  
public class ScannerExample {  
    public static void main(String args[]){  
        Scanner in = new Scanner(System.in);  
        System.out.println("Enter your name: ");  
        String name = in.nextLine();  
        System.out.println("Name is: " + name);  
        in.close();  
    }  
}
```

Output:

```
Enter your name: Ali  
Name is: Ali
```

Example code to Understanding Scanner Class Basics:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        // Create a Scanner object
        Scanner scanner = new Scanner(System.in);
        // Prompt the user to enter their name
        System.out.print("Enter your name: ");
        // Read the user's input as a String
        String name = scanner.nextLine();
        // Prompt the user to enter their age
        System.out.print("Enter your age: ");
        // Read the user's input as an integer
        int age = scanner.nextInt();

        // Display the user's name and age
        System.out.println("Hello, " + name + "! You are " + age + " years old.");

        // Close the scanner
        scanner.close();
    }
}
```

This code prompts the user to enter their name and age, reads the input using the Scanner class, and then displays a greeting along with the user's name and age.

Example code to Practice Input Reading:

```
import java.util.Scanner;
public class InputReading {
    public static void main(String[] args) {
        // Create a Scanner object to read input from the console
        Scanner scanner = new Scanner(System.in);

        // Read different types of input data
        System.out.print("Enter an integer: ");
        int intValue = scanner.nextInt();

        System.out.print("Enter a double: ");
        double doubleValue = scanner.nextDouble();

        scanner.nextLine(); // Consume the newline character left by nextDouble()
    }
}
```

```
System.out.print("Enter a string: ");
String stringValue = scanner.nextLine();

System.out.print("Enter a character: ");
char charValue = scanner.next().charAt(0); // Read the first character of input

// Display the input
System.out.println("Integer: " + intValue);
System.out.println("Double: " + doubleValue);
System.out.println("String: " + stringValue);
System.out.println("Character: " + charValue);

// Close the scanner
scanner.close();
}
```

Output:

```
Enter an integer: 205
Enter a double: 20.65
Enter a string: SSUET
Enter a character: A

Integer: 205
Double: 20.65
String: SSUET
Character: A
```

Exercise

- Write a program to make BIO_DATA and take several user inputs (i.e. name, age, gender, department, email id, father name) and print the result along with the greeting message.
- Write a program which takes two numbers as input from user and calculate sum and difference of both numbers.
- Take a three-digit number from user, separate each digit of the number and sum them For
example: 123
1+2+3= 6